Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

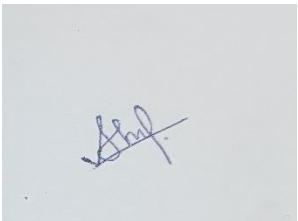
Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.



Srushti Chaudhari
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2025-26 (EVEN)**	**Course Name**	**AIML Lab**
Semester	**6**	**Course Code**	**23CT1602**
Roll No	**23**	**Name of Student**	**Srushti Chaudhari**

Practical Number	3
Course Outcome	CO1: Apply Artificial Intelligence problem formulation techniques, intelligent agents, search algorithms, and logical inference to solve real-world problems. CO2: Implement and evaluate machine learning algorithms including classification, clustering, and dimensionality reduction using appropriate preprocessing and performance metrics. CO3: Analyze and compare ensemble learning models.
Aim	Modeling a Real-World Problem as a Search Problem (8-Puzzle) To model the **8-Puzzle problem** using **state-space representation** and demonstrate how search techniques can be used to reach the goal state.
Problem Definition	The **8-Puzzle problem** consists of a **3×3 grid** containing eight numbered tiles and one blank space. The tiles can slide into the blank space to create new configurations. The objective is to transform an **initial configuration** into a **goal configuration** by moving tiles in the minimum number of steps. Each move involves sliding a tile into the blank space in one of four directions: **up, down, left, or right**, if possible. This problem is commonly used in Artificial Intelligence to illustrate **state-space search**, where each board configuration represents a state and moves represent transitions between states.
Theory (100 words)	The **8-Puzzle problem** is a well-known example used in Artificial Intelligence to demonstrate **state-space search techniques**. In this problem, each arrangement of tiles represents a **state**, and moving a tile into the blank space represents an **action**. The collection of all possible states forms the **state space**. The goal is to find a sequence of actions that transforms the **initial state** into the **goal state**. Search algorithms explore the state space to find the optimal path with the minimum cost. Each move has a cost of 1, and the total path cost represents the number of moves required to reach the goal configuration.

Procedure and Execution

(100 Words)

Algorithm:

1. Start the program.
2. Define the **initial state** of the puzzle.
3. Define the **goal state** configuration.
4. Identify possible actions (move blank **Up, Down, Left, Right**).
5. Generate new states by swapping the blank with adjacent tiles.
6. Check if the new state matches the goal state.
7. Continue generating states until the goal configuration is reached.
8. Display the sequence of moves and number of steps.
9. Stop the program.

Code:

```
initial_state = [1,2,3,
                 4,0,5,
                 6,7,8]

goal_state = [1,2,3,
              4,5,0,
              6,7,8]

print("Initial State:")
print(initial_state[0:3])
print(initial_state[3:6])
print(initial_state[6:9])

print("\nGoal State:")
print(goal_state[0:3])
print(goal_state[3:6])
print(goal_state[6:9])
```

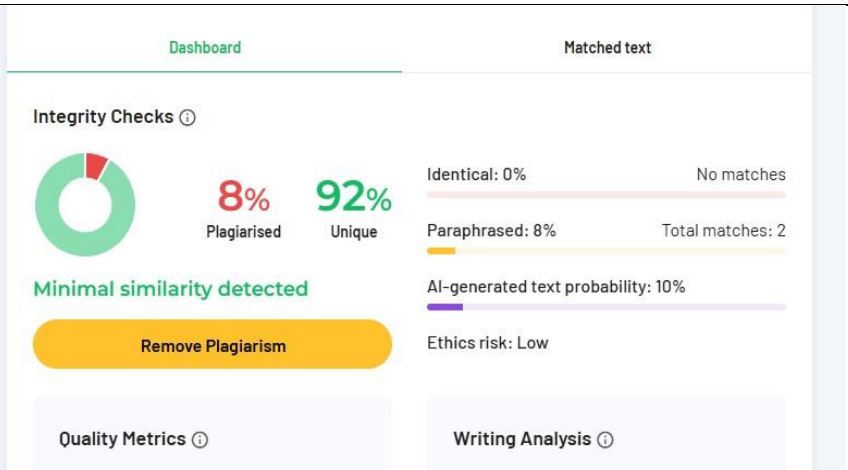
Output:

```
•••  Initial State:
     [1, 2, 3]
     [4, 0, 5]
     [6, 7, 8]

     Goal State:
     [1, 2, 3]
     [4, 5, 0]
     [6, 7, 8]
```

Output Analysis

The program represents the **initial state** and **goal state** of the 8-Puzzle problem. Each configuration represents a state in the state-space. By applying valid moves, the puzzle can transition from one state to another until the goal configuration is achieved.

Link of student Github profile where lab assignment has	https://github.com/SRchaudhari15/AIML-Practicals.git
Conclusion	The 8-Puzzle problem was successfully modeled using **state-space representation**. The experiment demonstrates how real-world problems can be represented as search problems where states, actions, and goal conditions help determine the solution path.
Plag Report (Similarity index < 12%)	
Date	[redacted] 29/12/25

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